

WEEK3 TASKS

1. PASSWORD CRACKING WITH JTR

John the Ripper, also known as JTR, is a password cracking tool that is used in cyber-security for checking password security. It can work with different types of password hashes and can also help recover passwords from protected files such as PDF, ZIP, and Office files. JTR is available for different operating systems including Linux, Windows, and Mac.

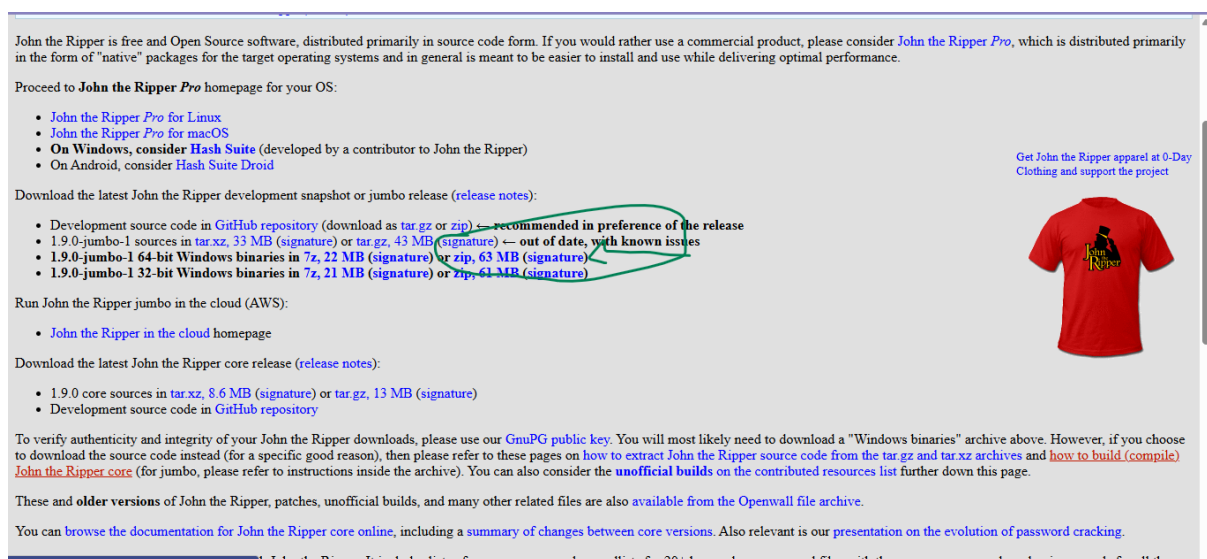
Johnny is the graphical version of John the Ripper. It provides an easy interface where we can use the tool by clicking different options instead of entering many commands in the terminal. This makes Johnny easier to use for beginners.

In this lab, we will use John the Ripper and Johnny to find the password of a protected PDF file. Through this activity, we will learn how password cracking works and understand why strong passwords are important for keeping files.

➤ STEP 1

Download John the Ripper from official website on your windows PC:

<https://www.openwall.com/john/>



The screenshot shows the official John the Ripper website. A green circle highlights the download instructions for Windows, specifically pointing to the '1.9.0-jumbo-1 64-bit Windows binaries in 7z, 22 MB (signature) or zip, 63 MB (signature)' option. A red t-shirt with the John the Ripper logo is visible on the right side of the page.

John the Ripper is free and Open Source software, distributed primarily in source code form. If you would rather use a commercial product, please consider [John the Ripper Pro](#), which is distributed primarily in the form of "native" packages for the target operating systems and in general is meant to be easier to install and use while delivering optimal performance.

Proceed to [John the Ripper Pro](#) homepage for your OS:

- [John the Ripper Pro](#) for Linux
- [John the Ripper Pro](#) for macOS
- **On Windows, consider [Hash Suite](#)** (developed by a contributor to John the Ripper)
- On Android, consider [Hash Suite Droid](#)

Download the latest John the Ripper development snapshot or jumbo release ([release notes](#)):

- Development source code in [GitHub repository](#) (download as [tar.gz](#) or [zip](#)) ← **recommended in preference of the release**
- 1.9.0-jumbo-1 sources in [tar.xz](#), 33 MB (signature) or [tar.gz](#), 43 MB (signature) ← **out of date, with known issues**
- 1.9.0-jumbo-1 64-bit Windows binaries in [7z](#), 22 MB (signature) or [zip](#), 63 MB (signature)
- 1.9.0-jumbo-1 32-bit Windows binaries in [7z](#), 21 MB (signature) or [zip](#), 61 MB (signature)

Run John the Ripper jumbo in the cloud (AWS):

- [John the Ripper in the cloud](#) homepage

Download the latest John the Ripper core release ([release notes](#)):

- 1.9.0 core sources in [tar.xz](#), 8.6 MB (signature) or [tar.gz](#), 13 MB (signature)
- Development source code in [GitHub repository](#)

To verify authenticity and integrity of your John the Ripper downloads, please use our [GnuPG public key](#). You will most likely need to download a "Windows binaries" archive above. However, if you choose to download the source code instead (for a specific good reason), then please refer to these pages on [how to extract John the Ripper source code from the tar.gz and tar.xz archives](#) and [how to build \(compile\) John the Ripper core](#) (for jumbo, please refer to instructions inside the archive). You can also consider the [unofficial builds](#) on the contributed resources list further down this page.

These and **older versions** of John the Ripper, patches, unofficial builds, and many other related files are also available from the [Openwall file archive](#).

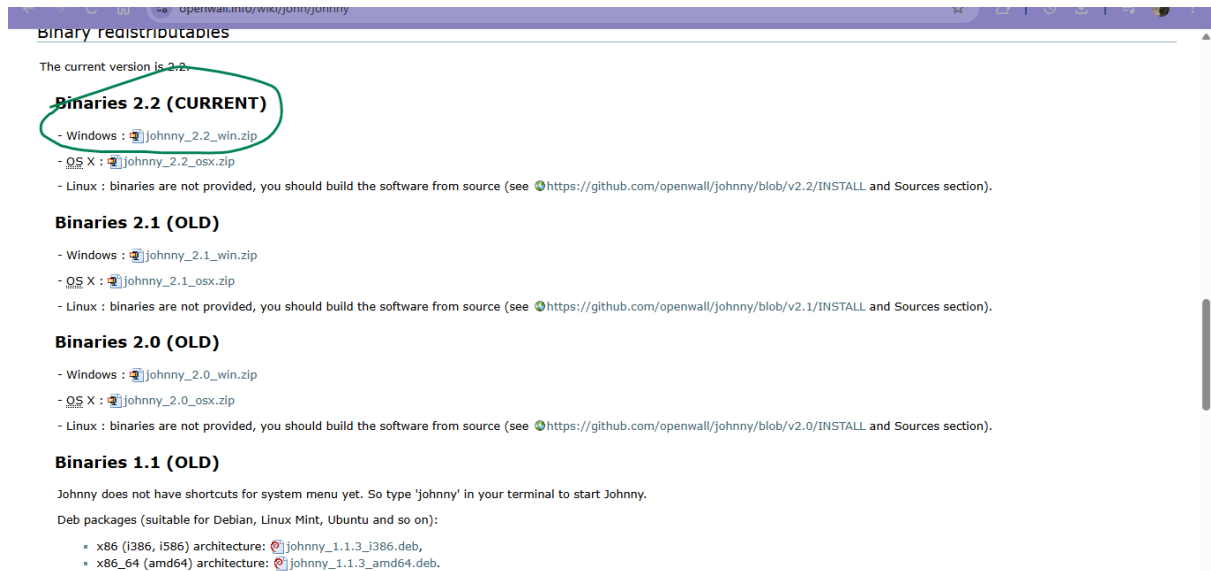
You can [browse the documentation](#) for John the Ripper core online, including a [summary of changes between core versions](#). Also relevant is our [presentation on the evolution of password cracking](#).

John the Ripper includes lists of common passwords, wordlists for 20+ human languages, and files with the common passwords and unique words for all the

➤ STEP 2

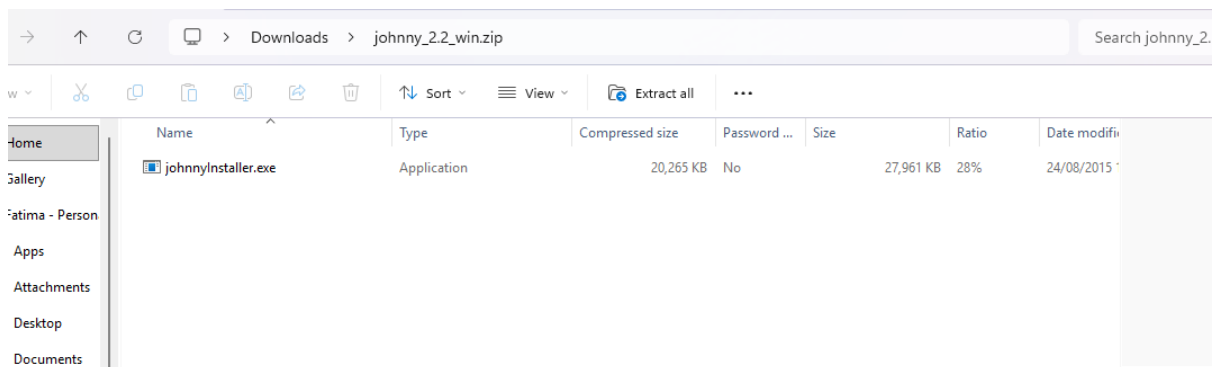
Download Johnny GUI from official website:

<https://openwall.info/wiki/john/johnny>



➤ STEP 3

Run the setup file & install Johnny as shown in below:

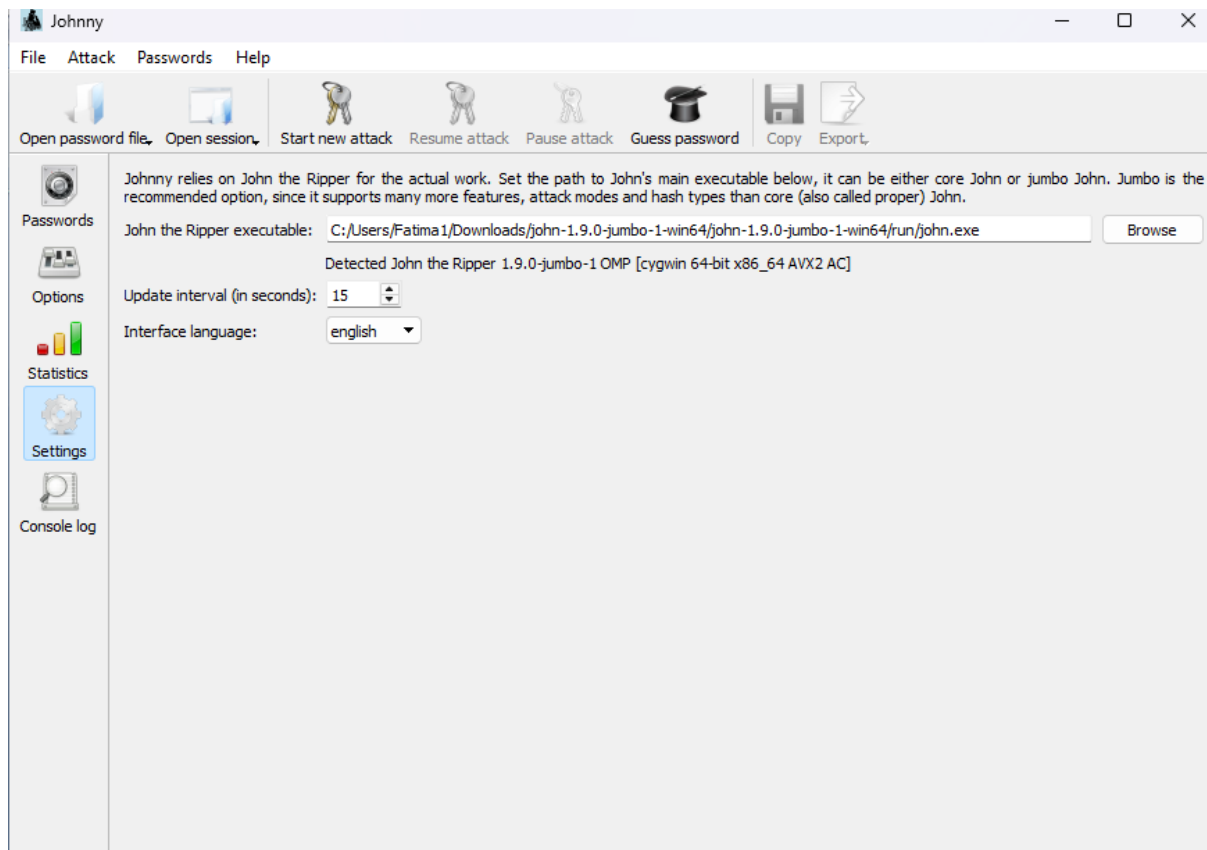


➤ STEP 4

After installation, open Johnny

➤ STEP 5

Click on settings & browse:

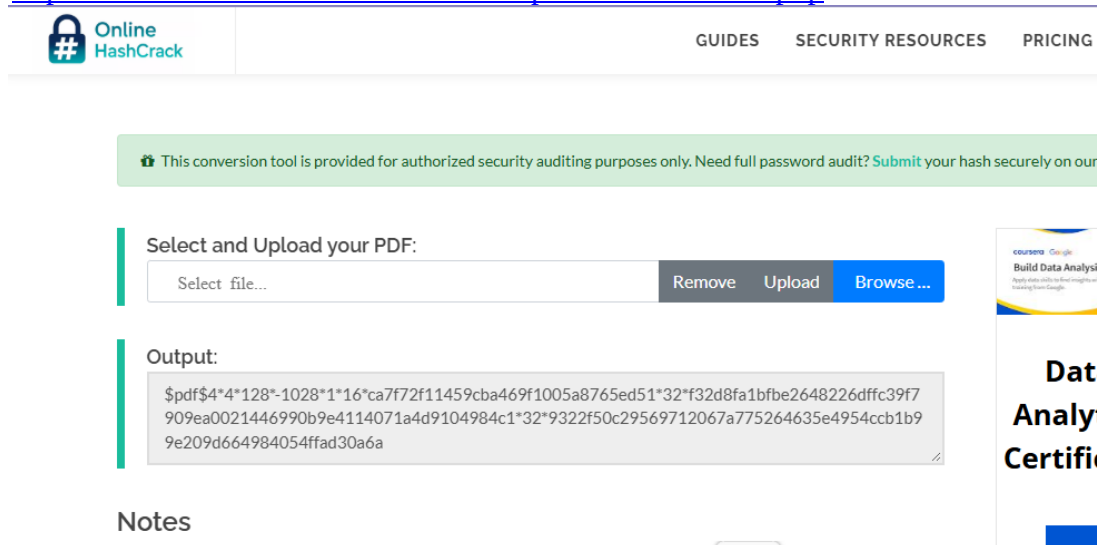


(you have to select john.exe and its present in run folder)

➤ STEP 6

Download the encrypted PDF file to your PC. And Open the hash website & upload your pdf file to find its hash:

<https://www.onlinehashcrack.com/tools-pdf-hash-extractor.php>



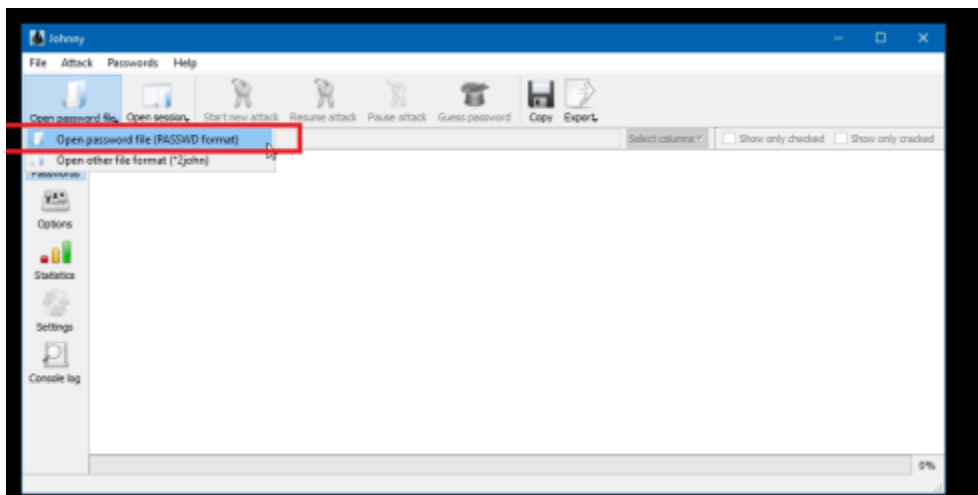
*Note: If your hash contains extra characters like b' in the start then make sure to remove those when saving in txt file (hash value should be in the format shown in above screenshot i.e. starting with \$pdf\$....

➤ STEP 7

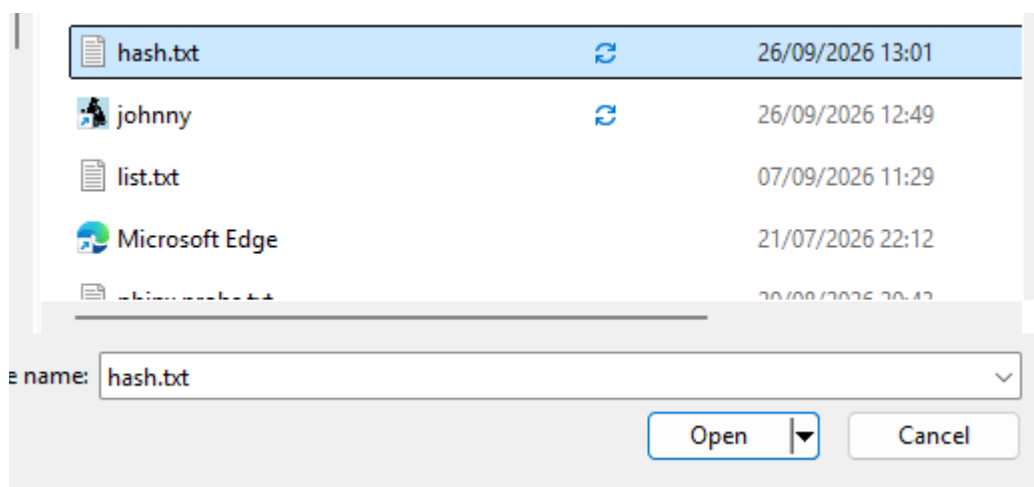
Open notepad and Paste the hash value inside notepad and save the file in txt format.

➤ STEP 8

Open Johnny again and Click on 'Open password file'

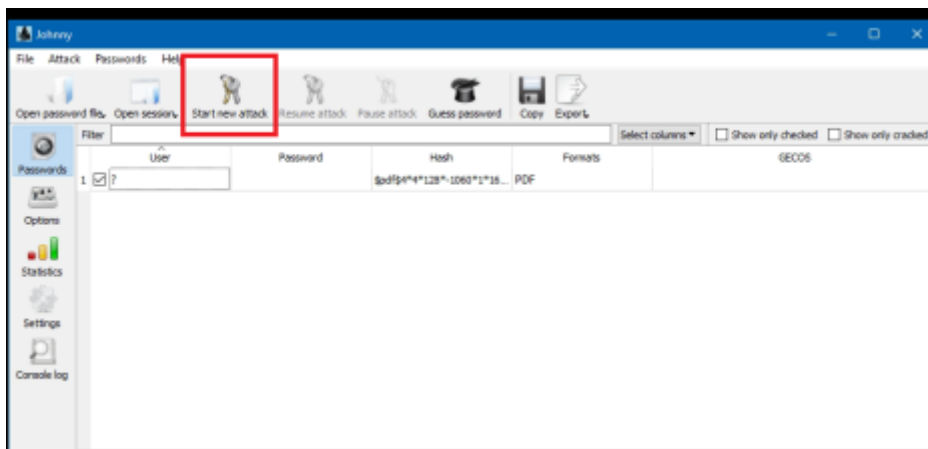


Browse to the hash1.txt file that you have just saved & click on Open:

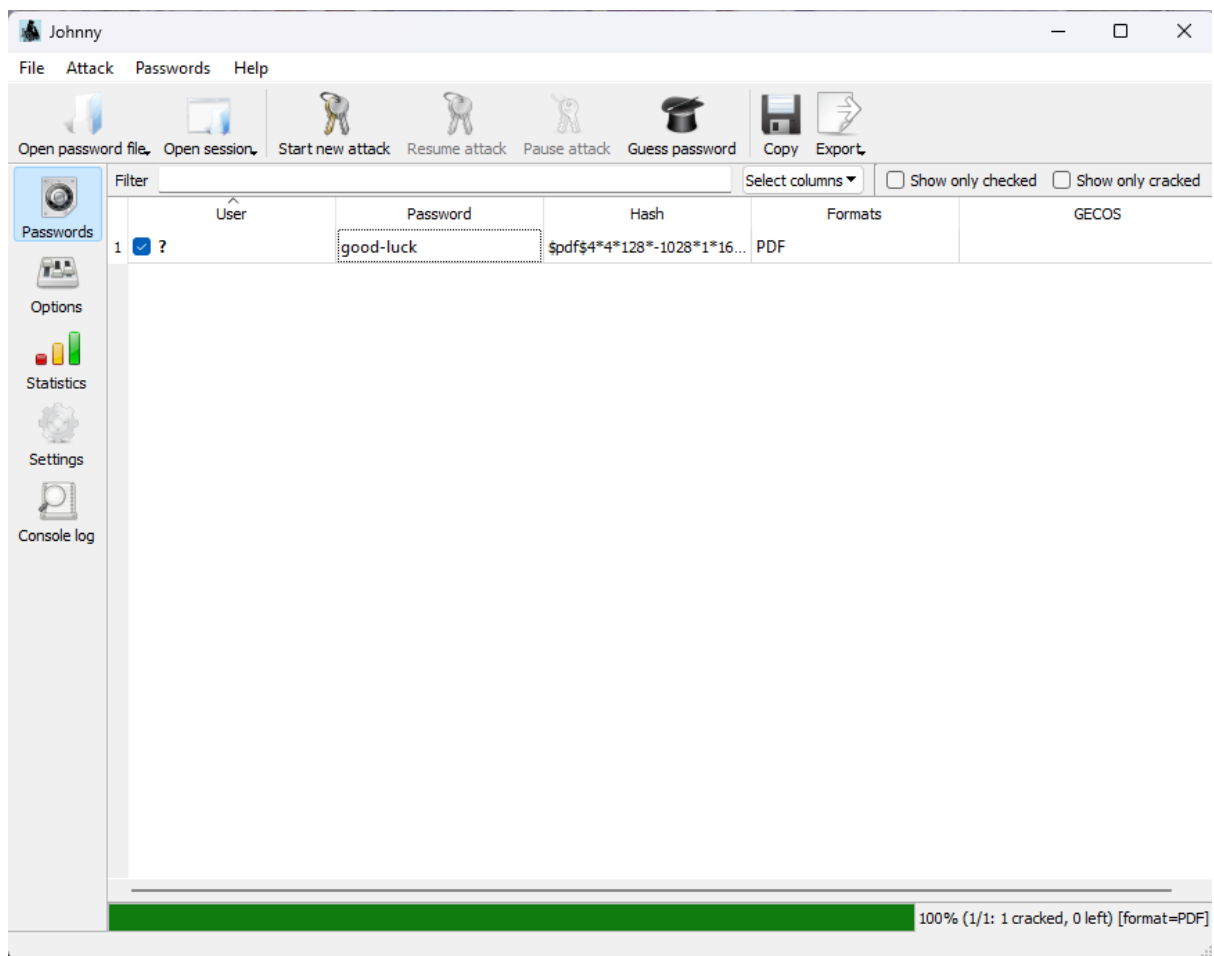


➤ STEP 9

Click on 'Start new attack':



Then the pass will show



Open the encrypted PDF and entre the password and enjoy!!